1. Scope and design basis

REVIEW COPY - NOT FOR SUBMISSION. The native drawing, airflow allocation and duct schedule are available. Maximum-load diffuser ADPI is unresolved. This report records the current design honestly and is generated from the same data as the drawing.

The assignment requires maximum and 40% airflow diffuser selection, characteristic length and ADPI checks, 4-5 diffusers per VAV box, flexible connections and a sized single-line duct network. This report documents those tasks; it does not assert that a presentation is required.

Input	Value	Evidence / status
Office load area	1022 m2	Printed source drawing label
External footprint	45 x 30 m	Native DWG geometry; units mm
Cooling allowance	120 W/m2 total	Preliminary assumption
Sensible heat ratio	0.75	Preliminary assumption
Room / supply air	24 / 12 deg C	Assumed; delta T = 12 K
Ceiling height	2.7 m	Assumed, requires coordination
Minimum airflow	40% of maximum	Assignment requirement
Flexible duct limit	4 m	FSD 2/2023 section 7.2
Rigid friction limit	0.8 Pa/m	Chosen design criterion
Main / branch velocity	5 / 4.5 m/s	Chosen design criteria

Rectangular envelope cells total 1042.10 m2 and include columns and small recesses. Their proportions allocate the printed 1022 m2 load area; this is not a claim of measured net room area. Originals remain preserved.

2. Cooling load and air balance

Total cooling = 1022 x 120 / 1000 = **122.64 kW**. Sensible cooling = 122.64 x 0.75 = **91.98 kW**. These are cooling loads, not electrical input.

Supply volume = sensible heat / (air density x specific heat x temperature difference). With density 1.2 kg/m3 and specific heat 1006 J/(kg K), Q = 91980 / (1.2 x 1006 x 12) = **6.3494 m3/s** = **6349.4 L/s**. Minimum supply = **2539.8 L/s**.

Each diffuser receives airflow proportional to its allocated load area. Each VAV box serves four outlets. The minimum load of 36 W/m2 sensible follows constant supply temperature and 40% airflow; it is an assumed operating model, not a measured building load.

VAV	Outlets	Maximum L/s	Minimum L/s
VAV-01	4	695.9	278.4
VAV-02	4	695.9	278.4
VAV-03	4	695.9	278.4
VAV-04	4	711.0	284.4
VAV-05	4	689.8	275.9
VAV-06	4	689.8	275.9
VAV-07	4	689.8	275.9
VAV-08	4	704.8	281.9
VAV-09	4	412.6	165.0
VAV-10	4	363.8	145.5

Outdoor-air requirement, return-air routing, fan selection and complete plant capacity are not proven by this supply-air balance. Occupancy and system information are needed before claiming full building-system compliance.

3. Diffuser selection and ADPI correction

Selected review arrangement: 40 Krueger 1400/51400 outlets. Lower zone: 600 mm panels with 8 in necks. Upper zone: 300 mm panels with 6 in necks. Throws are linearly interpolated within the published metric rows, without extrapolation; the third catalogue throw is T0.25.

Classification correction: the manufacturer engineering guide a1-11 identifies 1400 as Type 3 circular-pattern. The generic louvered 1.0-3.4 ratio screen in the original v01 CSV is retired as an acceptance test. For circular patterns, ASHRAE 2015 Table 3 uses distance to the closest wall or intersecting jet.

At 90 W/m2 sensible, the current maximum-flow ratios fall outside both the 60 W/m2 range 0.7-1.3 and the 125 W/m2 range 0.5-1.5 in ASHRAE 2015 Table 4. Neither table gives a tested 90 W/m2 row. This is evidence to revise the design, not an exact prediction of its ADPI.

At minimum load, 36 W/m2 lies between the 2019 round-diffuser 25 and 50 W/m2 cases. Comparing both tabulated envelopes is diagnostic; no measured ADPI percentage is invented. K-Select could supply more operating-point data, but its installer component was blocked by Windows Security.

Example	Q max/min L/s	Closest L m	T/L max/min
SD-01	170.8 / 68.3	2.40	1.95 / 1.23
SD-08	178.3 / 71.3	2.40	2.00 / 1.26
SD-32	175.3 / 70.1	2.40	1.98 / 1.24
SD-33	125.6 / 50.2	1.80	2.56 / 1.62
SD-40	73.6 / 29.5	1.50	2.33 / 1.33

All 40 detailed rows are in adpi_classification_audit_v01.json and diffusers_v01.csv. Current release status remains NOT ACCEPTED. Geometry, neck size and operating assumptions must be reconsidered with applicable performance evidence before final issue.

4. VAV layout and flexible connections

Ten VAV boxes serve four outlets each. Their 1000 x 800 mm plan envelopes are schematic allowances, not manufacturer-selected equipment dimensions. Flexible routes run directly from box corners to outlets; the calculated route adds 0.5 m for connection/drop allowance.

VAV	Longest design flex m	Connections	Limit check
VAV-01	3.473	4	Within 4 m
VAV-02	3.473	4	Within 4 m
VAV-03	3.473	4	Within 4 m
VAV-04	3.473	4	Within 4 m
VAV-05	3.473	4	Within 4 m
VAV-06	3.473	4	Within 4 m
VAV-07	3.473	4	Within 4 m
VAV-08	3.473	4	Within 4 m
VAV-09	2.640	4	Within 4 m
VAV-10	2.472	4	Within 4 m

Maximum flexible design length is 3.473 m. Lower-zone flex is nominal 250 mm; upper-zone flex is 200 mm. Product-specific flexible-duct pressure losses, actual bends and reducer losses are not included in the straight rigid-duct friction schedule.

FSD Circular 2/2023 section 7.1 specifies flexible-duct material/test requirements and section 7.2 normally limits length to 4 m. Fire-compartment penetrations and the AHU/corridor crossing require a coordinated fire strategy. Section 9.2 protected-area requirements cannot be satisfied merely by drawing a fire-damper symbol.

5. Rigid duct sizing schedule

Rectangular velocity uses actual area: v = Q/(W x H). Equal-friction equivalent diameter: De = 1.30(WH)^0.625/(W+H)^0.25. Darcy-Weisbach uses standard-air density 1.204 kg/m3, viscosity 1.81e-5 Pa s and assumed roughness 0.09 mm. Sizes use 50 mm increments and aspect ratio no greater than 4.

The reviewed MIT Serghide correlation from python-hvac is checked against iterative Colebrook and five ASHRAE Table 13 points. These checks validate the implementation; they do not replace fitting-loss or full fan-pressure calculations.

Section	L/s	W x H mm	m/s	Pa/m
M01	6349.4	1500 x 850	4.98	0.207
M02	2771.5	800 x 700	4.95	0.320
M03	1385.8	700 x 400	4.95	0.509
M04	3577.9	1200 x 600	4.97	0.298
M05	2192.1	800 x 550	4.98	0.382
B01A	1385.8	700 x 450	4.40	0.374
B01B	695.9	450 x 350	4.42	0.563
B01C	695.9	450 x 350	4.42	0.563
B01D	695.9	450 x 350	4.42	0.563
B01E	695.9	450 x 350	4.42	0.563
B02A	1385.8	700 x 450	4.40	0.374
B02B	695.9	450 x 350	4.42	0.563
B02C	695.9	450 x 350	4.42	0.563
B02D	695.9	450 x 350	4.42	0.563
B02E	695.9	450 x 350	4.42	0.563
B03A	1385.8	700 x 450	4.40	0.374
B03B	695.9	450 x 350	4.42	0.563
B03C	695.9	450 x 350	4.42	0.563
B03D	695.9	450 x 350	4.42	0.563
B03E	695.9	450 x 350	4.42	0.563
B04A	1415.7	700 x 450	4.49	0.389
B04B	711.0	400 x 400	4.44	0.558
B04C	711.0	400 x 400	4.44	0.558
B04D	711.0	400 x 400	4.44	0.558
B04E	711.0	400 x 400	4.44	0.558
M06	776.4	450 x 350	4.93	0.689
B09A	412.6	400 x 250	4.13	0.670
B09B	412.6	400 x 250	4.13	0.670
B10A	363.8	350 x 250	4.16	0.726
B10B	363.8	350 x 250	4.16	0.726

6. Verification and reference trail

Completed evidence: native AutoCAD 2027 source inspection (INSUNITS=4); a three-diffuser placement/plot sample; native review DWG save and reopen; review PDF inspection; 40 outlets / 10 VAVs / 40 flex routes; source hashes preserved; reproducible air balance and straight-duct sizing.

Remaining acceptance work: resolve maximum/minimum diffuser ADPI with the correct pattern and applicable load evidence; revise layout/selection as necessary; label all short duct sections legibly; complete friction-chart evidence; regenerate and inspect final drawing sheets; reconcile every final schedule and drawing count. No final release is claimed.

EMSD BEC 2024 Table 6.4 is a load-design-condition reference, not a prescribed W/m2 allowance. VAV fan power and static-pressure control provisions require system evidence. BEC water-pipe friction provisions are not air-duct friction limits.

[1] BSE3227 Mini design assignment, pp. 1-2, tasks 1-4; supplied Project layout.dwg and Layout.pdf.

[2] Krueger 1400/51400 published performance, B1-11/B1-12; Air Distribution Engineering a1-10/a1-11; Reference Charts B1-10. Official downloads are preserved in 02_References.

[3] ASHRAE Applications 2015, Chapter 57 Tables 3-4; Applications 2019, Chapter 58 Tables 5-6A. Official handbook HTML snapshots are preserved.

[4] ASHRAE Fundamentals 2021, Chapter 21, Figure 10, Equation 25, Tables 12-13.

[5] EMSD Building Energy Code 2024, Table 6.4; sections 6.7.2-6.7.5 and 6.10.6.1-2.

[6] Hong Kong FSD Circular 2/2023, Mechanical Ventilating Systems: sections 3, 7.1-7.2 and 9.2.

[7] Liu, Clark and Novoselac (2017), Energy and Buildings 134, 271-284. The supplied paper tests 25/50 W/m2; it does not prescribe this office load.

[8] TomLXXVI/python-hvac, MIT, pinned commit 94247560a667905bfbf67e45a58965a58254fce8. See 06_Tools/TOOL_REVIEW.md and preserved licence.

Exact URLs, source locators and assumptions: 02_References/SOURCE_REGISTER.md. Calculation inputs and audit records: 05_Calculations/data. This report is reproducible with build_report.py.